



# Jukka Niiranen

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## Governance as Code



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# Agenda

- The code phenomenon
- Pre-code era governance tooling
- Doing it the code way
- DEMOS!



Lun@Lake



Fellowwind



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# Hi, I'm Jukka

## I write things on the internet.

...about governance, licensing, low code, citizen development, AI hype, software business, CRM, UX, enshittification, digital sovereignty, among other things.



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[perspectives.plus](https://perspectives.plus) - [licensing.guide](https://licensing.guide) - [niiranenadvisory.com](https://niiranenadvisory.com)



2012

“Everyone should  
learn how to code.”

2026

“Everyone should  
learn about code.”

# Vibe check: what's your coder profile? 🙋



## A) Pro coder

- Knows how to write production-ready code without any AI help.



## B) Vibe coder

- Relies on AI coding agents to write all code for them.

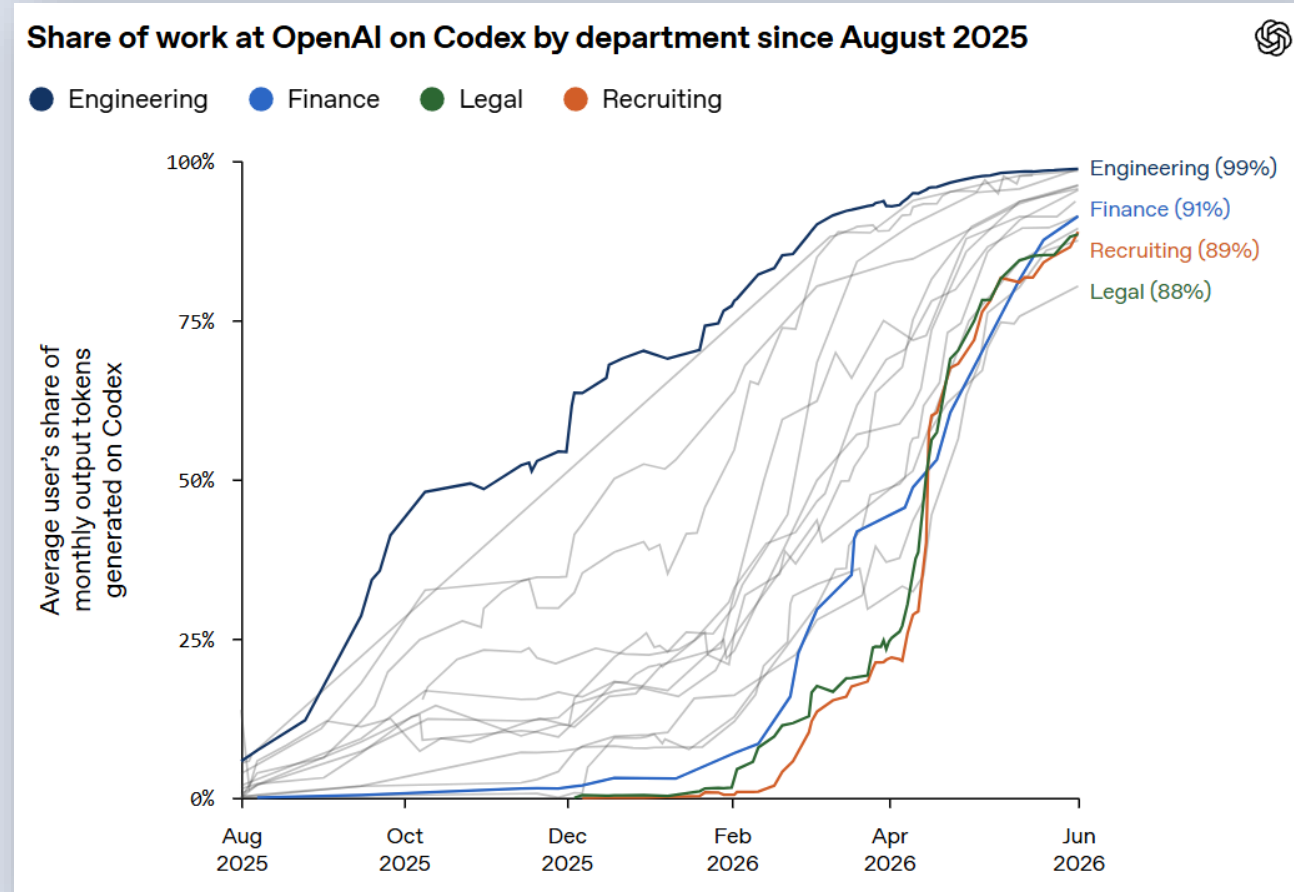


## C) Low/no coder

- Doesn't touch code, leverages GUI tools for building solutions.

# Code is the real UI for AI

- LLMs didn't suddenly become an all-knowing sentient entity. They were given a computer with deterministic tools to operate. And that's how agentic AI became useful.
- When code can be used in solving the user's problem, generic coding agents are now beating bespoke software products across the information worker spectrum – far beyond just software development tasks.



*"Non-developers are the fastest-growing user groups for Codex."*

<https://openai.com/index/how-agents-are-transforming-work/>

# From low-code to all-code platform

- Microsoft's early attempts with Copilot sidecars in GUI-based editors failed to deliver much value.
- Now, coding agents with plugins like [power-platform-skills](#) can easily build Power Platform artifacts that have proper support for API/CLI path.
- Everything that does NOT have code-first support remains a struggle for AI. Such as;
  - Model-driven apps: full maker portal code shipped alongside skill file.
  - Canvas apps: co-authoring MCP bridges a live browser session for agent to operate in.



Jul 24, 2026 • 13 min read

## The all-code Power Platform

Low-code application development isn't capturing the true benefits from AI. Maybe we should talk about "all-code development platforms" instead?



Jukka Niiranen

[Link](#)

# In loving memory of CoE Starter Kit

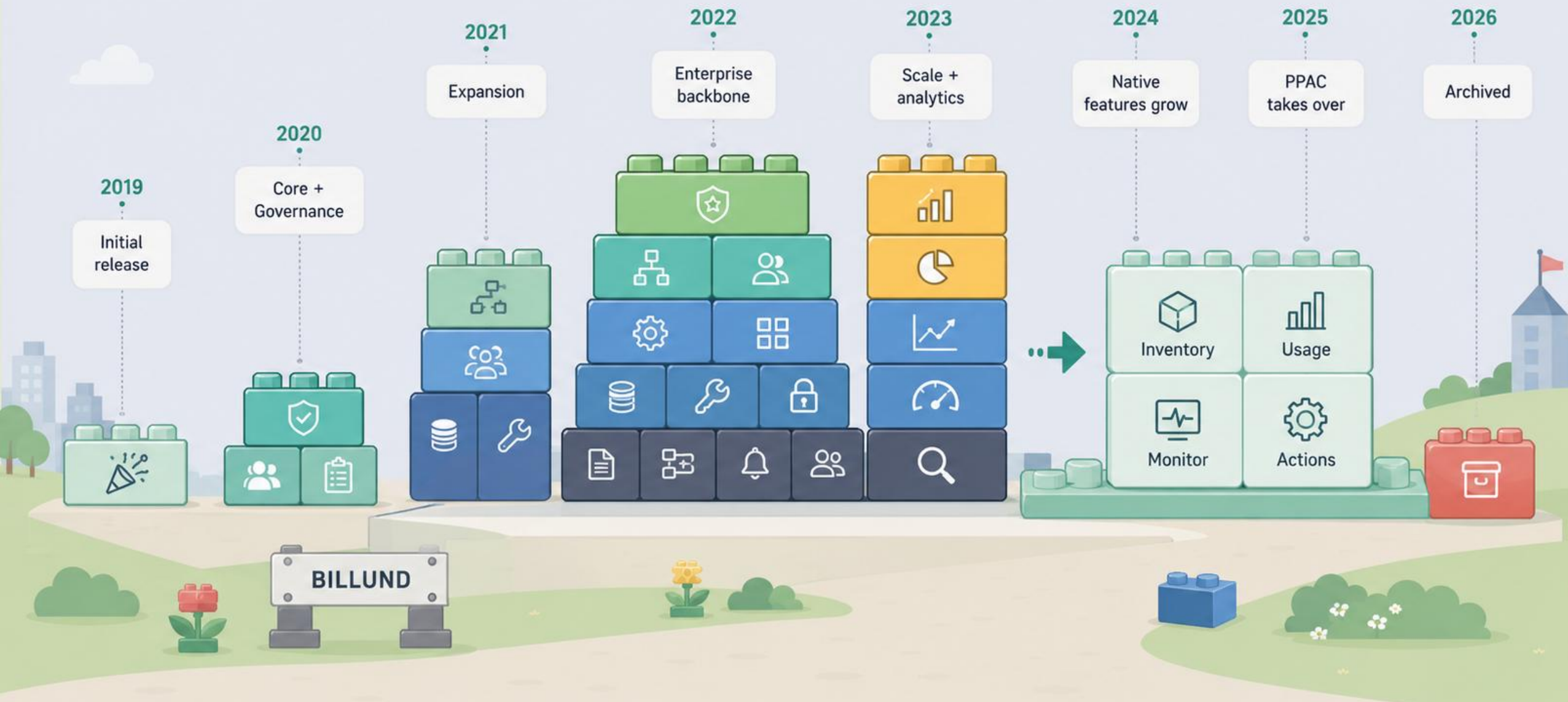


The screenshot shows the GitHub repository page for 'microsoft / coe-starter-kit'. The repository is marked as archived with a yellow banner stating: 'This repository was archived by the owner on Jul 2, 2026. It is now read-only.' The repository is a 'Public archive' with 75 unwatchers, 241 forks, and 946 stars. The main branch is 'main'. A search bar for files is present. The repository was last updated 5 years ago. The commit history shows a commit by Grant Archibald (Grant-Archibald-MS) 4 months ago with 1,329 commits. The releases section shows 66 releases, with the latest being 'CoE Starter Kit February 2026' from 7 months ago. Other links include '+ 65 releases', 'Readme', and 'License'.

# CoE Starter Kit

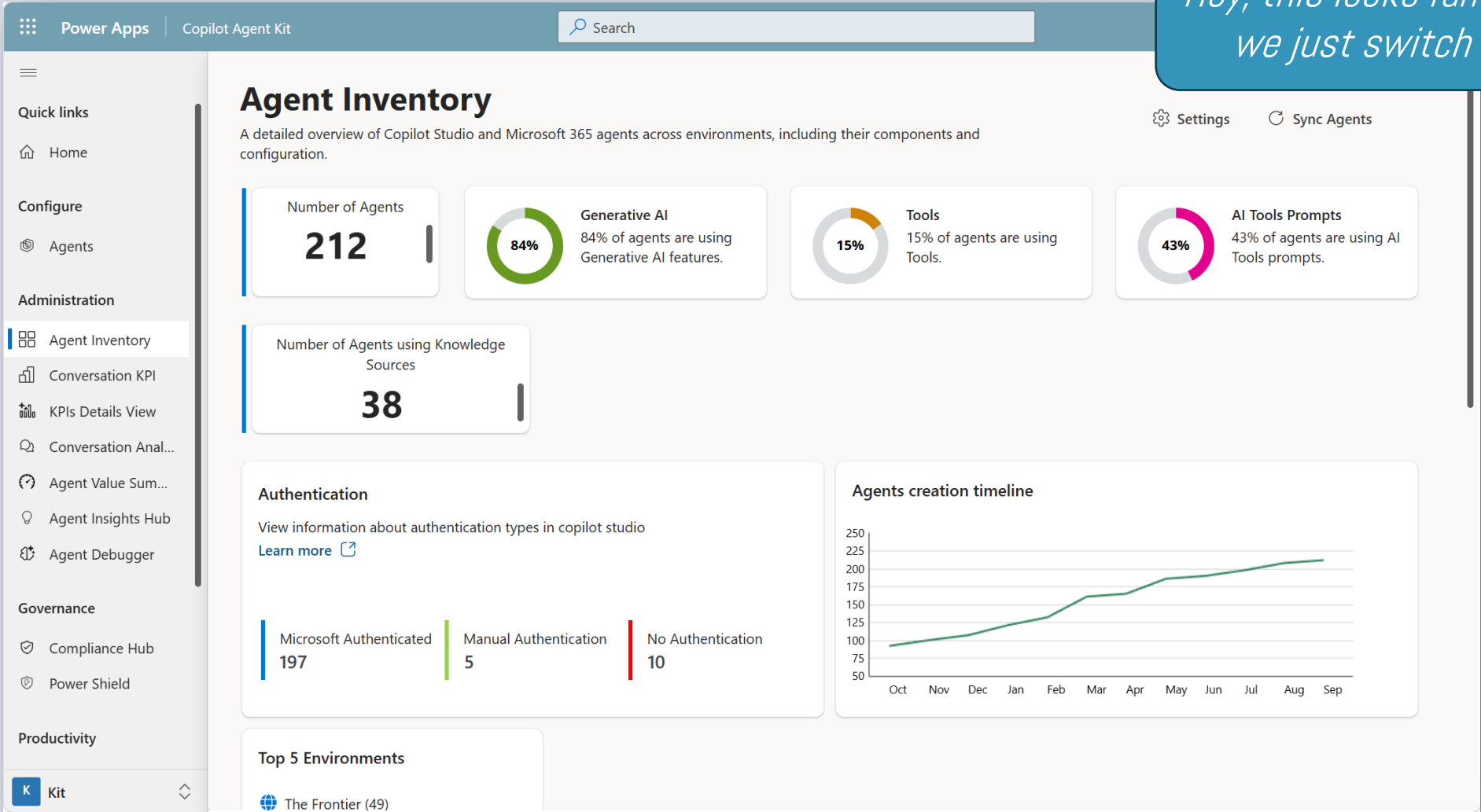
2019–2026

Evolution



# What about the Copilot Studio/Agent Kit?

Hey, this looks familiar! Could we just switch to this?



# What about the Copilot Studio/Agent Kit?

Ah, the good ol' sync flows activation experience...

Power Apps

Copilot Agent Kit

Search

Quick links

Home

Configure

Agents

Administration

Agent Inventory

Conversation KPI

KPIs Details View

Conversation Anal...

Agent Value Sum...

Agent Insights Hub

Agent Debugger

Governance

Compliance Hub

Power Shield

Productivity

Agent Library

Kit

You have 13 notifications. Select to view.

A setup wizard provides a guided step by step experience to configure the Copilot Agent Kit.

Confirm pre-requisites

Connection References

Configure Environment Variables

Activate Flows

Set up Agent Review

Finish

Activate Flows

Before using the features, the required flows must be activated. Turn on the GrandChild, then Child, and then rest of the flows in that order. Monitor each flow's status to ensure successful activation.

|                                                                      |            |     |         |
|----------------------------------------------------------------------|------------|-----|---------|
| Agent Insights   On Demand                                           | Cloud Flow | off | Turn On |
| Agent Insights   Pull Agent Metrics (Child)                          | Cloud Flow | off | Turn On |
| Agent Insights   Pull Transcript Metrics Per Date (Great-Grandchild) | Cloud Flow | off |         |
| Agent Insights   Scheduler                                           | Cloud Flow | off | Turn On |
| Agent Insights   Transcript Metrics Date Range (Grandchild)          | Cloud Flow | off | Turn On |
| Agent Inventory   Agent Actions                                      | Cloud Flow | off | Turn On |
| Agent Inventory   Agent Data for Actions                             | Cloud Flow | off | Turn On |

Back

Next

nordic summit

www.nordicsummit.info

#NordicSummit

# What about the Copilot Studio/Agent Kit?

The screenshot displays the Copilot Studio interface. On the left, a sidebar contains navigation links for Home, Agents, and Workflows. The main area is titled 'Review Agent Components' and includes tabs for Build, Activity, and Monitor. A central panel shows a list of components to add, including Agent, Classify, M365 Copilot, Human review, Connector, Function, Variable, If/Else, Loop, and Note. The workflow canvas on the right features a sequence of steps: 'Add a new row 4', 'Update a row 5', 'Add a new row', and 'Update a row', which all lead into a junction point. From this junction, the workflow branches into three paths: 'Update a row 4', 'Agent 2', and 'Agent 2 2'. A blue box next to 'Agent 2' provides a description: 'Agent 2 — Cross-skill reviewer. Reviews the complete skill set for overlap, redundancy, contradictions, coverage gaps, coupling, and consistency issues.' A prominent orange warning banner at the top of the canvas states: 'You've reached your credit limit. This published workflow will stop running until you request more credits from your admin. [Learn more about credits](#)'.

*...Now with also Copilot Studio credits requirement*

# Yeah, about those Managed Environments...

- The solution to a different problem than what CoE solved.
- It's great to have more built-in rules to configure what makers can & can't do in environments.
- Yet positioning Managed Environments as an in-product replacement for CoE Starter Kit in offering visibility into maker activity was a terrible idea.

Power Platform admin

Select Solution Checker rules to exclude from enforcement.

Usage insights

Get adoption insights, like top apps and flows, in this environment. [Learn more](#)

☒ Include insights for this environment in the weekly email digest

☐ Add additional recipients for the weekly email digest

Maker welcome content

Provide customized welcome content to help makers get started with this environment. [Learn more](#)

You are seeing this welcome content because this is a Managed Environment. It's cool is that! Not very? Well, yeah. True. I just wanted to type something in this box. Sorry for disturbing you. I'll let you get to the environment now.

1266/1500 characters

☐ Include consent button and log maker activity

Terms and conditions

<https://learn.microsoft.com/en-us/power-platform/admin/welcome-to-managed-environments>

[See preview](#) [Reset form](#)

Enable AI-generated app descriptions (preview)

Let Power Apps do the work for you. If descriptions of apps are blank, you can ask Power Apps to generate them. This will happen when an app is published and can help makers understand what the purpose of an app is.

☒ AI-generated app descriptions

[Save](#) [Close](#)

## Weekly update for the week of September 14, 2026

You're receiving this email because you're the admin for one or more managed environments.

### Key insights

Here's what happened in the 8 Managed Environments for tenant [redacted] in the last month:

|             |              |                  |
|-------------|--------------|------------------|
| 4           | 3            | 1                |
| Active apps | Active flows | Active app users |

### Consider cleaning these up

These apps haven't run in a while. For organizational hygiene and security, consider contacting the owners to update or remove them.

| App                                     | Environment | Last launch* |
|-----------------------------------------|-------------|--------------|
| <a href="#">Cleanup Old Objects App</a> | CoE         | None         |

| Flow                                                                      | Environment | Last launch* |
|---------------------------------------------------------------------------|-------------|--------------|
| <a href="#">Microsoft Teams Admin   Ask for Business Justification wh</a> | CoE         | None         |

### Most popular

These are the most-used canvas apps and flows of the last month.

| App                         | Environment   | Sessions |
|-----------------------------|---------------|----------|
| <a href="#">Advisory OS</a> | NA Production | 4        |

**There are only 3 flows in your Managed Environments**

| Flow                                                                    | Environment            | Runs |
|-------------------------------------------------------------------------|------------------------|------|
| <a href="#">Conversation KPI   Generate Conversation KPIs Scheduler</a> | Jukka's US Preview Dev | 54   |

| Maker          | Total Sessions | # Apps |
|----------------|----------------|--------|
| Jukka Niiranen | 5              | 3      |

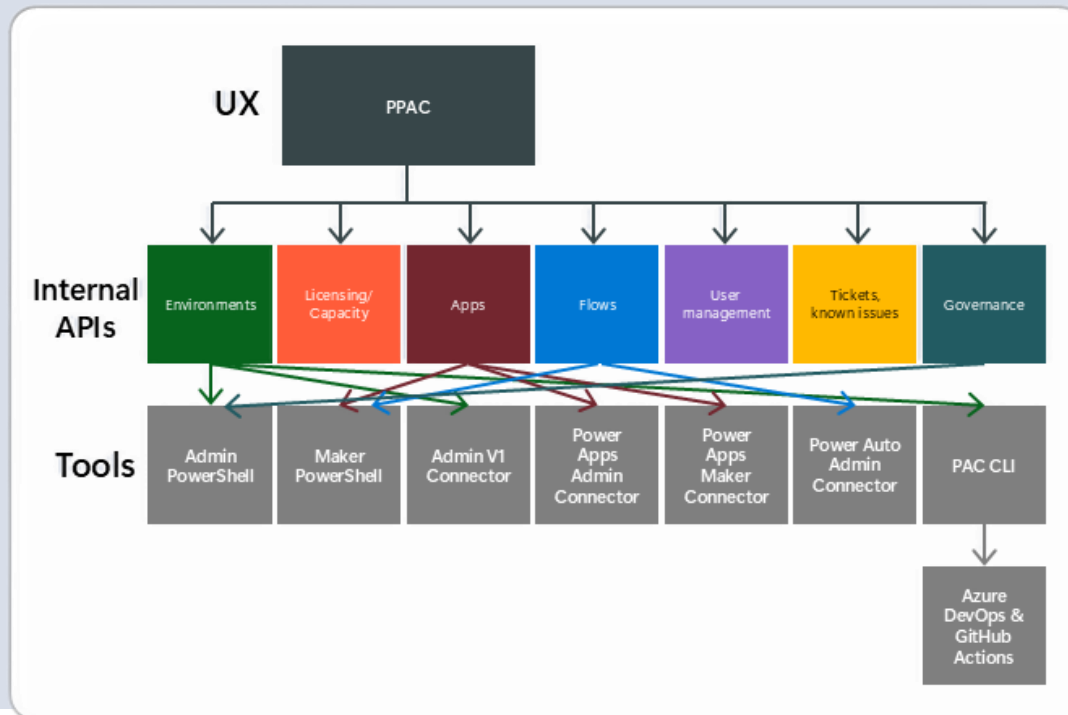
Get more visibility into your 11 other environments.

[Add Environment Management to more environments >](#)

# Power Platform API and SDKs: From UX-First to API-First (2025)

"Historically, Power Platform has empowered administrators through the Power Platform Admin Center (PPAC). This UX-first experience offered a seamless, intuitive interface for managing environments, automating everyday tasks, and discovering new capabilities within the platform.

Today, we're taking a bold step forward. We're transforming Power Platform into an API-first ecosystem—where every feature in PPAC is backed by a well-documented, publicly accessible API."



[Power Platform Dev Blog, July 2025](#)

# The Power Platform governance dilemma

- 😊 PPAC has grown and the features have matured.
- 🤔 ...Yet the built-in product features can't truly replace what CoE Starter Kit offered in terms of customizability.
- 🧐 APIs are more comprehensive than ever for governance tasks.
- 😬 ...But they are “invisible” unless you've proactively explored how to use them for tasks in the real world.

CODE

GUI



# Doing it the code way



# PPAC is not your only tool now

- The code tabs in the Microsoft docs were always there. Now, they are relevant to everyone.
- Example from [MS Learn](#): Manage Copilot Credits allocations programmatically.
- GUI is great for seeing what options and data are available. It's terrible for repeatedly performing the same actions.

## Manage Copilot credit allocations

This section shows the same end-to-end flow—check what's available, allocate credits to an environment, and control whether the environment draws from tenant capacity—in the Power Platform admin center and in each SDK. Use the tabs to switch between the admin center and your preferred language.

Power Platform admin center

PowerShell

C#

Python

1. Sign in to the [Power Platform admin center](#).
2. In the navigation pane, select **Licensing**.
3. Under **Products**, select **Copilot Studio**, and then select **Manage Copilot Credits**.

The screenshot displays the Power Platform admin center interface. The left navigation pane shows the 'Licensing' section selected. The main content area is titled 'Licenses' and includes a 'Summary' tab. Under 'Summary', there is a 'Copilot Studio' section with a 'Manage Copilot Credits' link. Below this, there are 'Recommendations' and a 'Capacity summary' section. The 'Capacity summary' section shows 'Pay-as-you-go Copilot Credits' with a value of 0 and 'Total Copilot Credits' with a value of 0. There is also a 'Prepaid capacity' section with a table showing capacity details.

| License type      | Category                 | Purchased | Assigned | Consumed |
|-------------------|--------------------------|-----------|----------|----------|
| Capacity          | Copilot Credits capacity | 0         | 0        | 0        |
| Capacity (Legacy) | Sessions capacity        | 0         | 0        | 0        |

# You have already been using the Inventory API

- PPAC inventory page shows a filtered version of the raw API data.
- Live inventory with ~15 min delay for new items in the tenant to show up.
- It's a very basic experience: free text search, column filter/sort, export to CSV and... that's it.

Power Platform admin center

Search for settings, pages, and more

**Manage**

- Home
- Actions
- Manage**
- Security
- Copilot
- Monitor
- Deploy...
- Licensing
- Support
- Admin centers

**Manage**

- Environments
- Environment groups
- Inventory**
- Usage
- Tenant settings
- Data
  - Data (preview)
  - Data integration
  - Data export
- Products
  - Power Apps
  - Power Automate
  - Power Pages
  - Dynamics 365 apps
  - Dynamics 365 agents
  - Copilot Studio
  - Dynamics 365 Commerce
  - Dataverse

**Maker inventory**

Oversee everything people make in the environments you manage. [Learn more](#)

**New columns available** Select the "Add and remove columns" icon to customize your view.

Refresh

Showing 853 of 853 items

| Item name                                  | Item type        | Modified on | Connectors    |
|--------------------------------------------|------------------|-------------|---------------|
| <a href="#">Report Builder</a>             | Agent            | —           | —             |
| <a href="#">Report Builder</a>             | Agent            | —           | —             |
| <a href="#">CoE Inventory Agent</a>        | Agent            | —           | shared_common |
| <a href="#">Solution Health Hub</a>        | Model-driven app | 09/08/26    | —             |
| <a href="#">Power Pages Managen</a>        | Model-driven app | 09/08/26    | —             |
| <a href="#">Power Platform Environ</a>     | Model-driven app | 09/08/26    | —             |
| <a href="#">Authentication &amp; Anti-</a> | Model-driven app | 09/08/26    | —             |
| <a href="#">Authentication &amp; Anti-</a> | Model-driven app | 09/07/26    | —             |
| <a href="#">Sales Close Agent</a>          | Agent            | —           | —             |
| <a href="#">Power Platform Contro</a>      | Code app         | 09/03/26    | —             |
| <a href="#">Sales Close Agent</a>          | Agent            | —           | —             |

# Azure resource graph explorer for *PowerPlatformResources*

The screenshot shows the Microsoft Azure Resource Manager (ARM) Resource graph explorer interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and the Copilot icon. The main header shows 'Home > Resource Manager' and 'Resource Manager | Resource graph explorer' for the 'Niiranen Advisory Oy' directory. A search bar is present, and a query is being executed: 'New query 2 \*'. The query is a Kusto query (KQL) that filters for 'PowerPlatformResources' and summarizes them by environment name.

```
1 PowerPlatformResources
2 | where type !~ "microsoft.powerplatform/environments"
3 | extend environmentId = tostring(properties.environmentId)
4 | join kind=leftouter (
5   PowerPlatformResources
6   | where type =~ "microsoft.powerplatform/environments"
7   | project environmentId = name, envName = tostring(properties.displayName)
8 ) on environmentId
9 | summarize resources = count() by envName
10 | order by resources desc
```

The results are displayed in a table with columns: envName, resources, and Details. The table shows the number of resources for each environment.

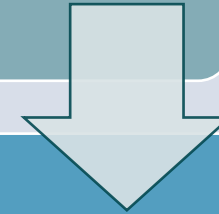
| envName            | resources | Details                      |
|--------------------|-----------|------------------------------|
| CoE                | 250       | <a href="#">View details</a> |
| Jukka PAYG         | 88        | <a href="#">View details</a> |
| Dynamics 365 Demos | 79        | <a href="#">View details</a> |
| Default Dumpyard   | 74        | <a href="#">View details</a> |
| The Frontier       | 67        | <a href="#">View details</a> |

- You can run custom Kusto queries (KQL) on the inventory data live in the Azure portal's ARG explorer.
- Very flexible for ad-hoc data retrieval.
- Very basic for results visualization, you have to build dashboards with something on top of this.

# The low-coder's journey into all-code Power Platform governance

## 1. Open the terminal, run PAC CLI

- Learn to access governance data



## 2. Clone a community repo, NPM run dev

- Learn to visualize and analyze governance data



## 3. Vibe code your own solution with GitHub Copilot / Claude / Codex /...

- Learn to manage and automate governance data

# Stage 1: PAC CLI

# Getting your PAC CLI ready

| Task                           | Command                                                                | What it does                                        |
|--------------------------------|------------------------------------------------------------------------|-----------------------------------------------------|
| 1. Check .NET                  | <code>dotnet --version</code>                                          | Confirms the prerequisite is installed              |
| 2A. Install PAC CLI            | <code>dotnet tool install --global Microsoft.PowerApps.CLI.Tool</code> | Installs PAC CLI for your Windows user              |
| 2B. Update PAC CLI             | <code>dotnet tool update --global Microsoft.PowerApps.CLI.Tool</code>  | Updates an existing installation                    |
| 3. Check PAC                   | <code>pac</code>                                                       | Shows that PAC runs and displays its version        |
| 4. Sign in                     | <code>pac auth create</code>                                           | Creates a local authentication profile              |
| 5. Check the active tenant/env | <code>pac auth list</code><br><code>pac auth who</code>                | Shows the active profile, user and organization     |
| 6. Switch tenant if needed     | <code>pac auth select --index 2 (example)</code>                       | Selects and confirms the intended profile           |
| 7. Run a safe command          | <code>pac env list</code>                                              | Lists environments you can access; makes no changes |

# PAC 101

- 'pac' : shows all the command groups
- 'pac [group] [command] [parameters]'
- Traditionally used for dev and config tasks, now expanding to allow broad querying of tenant resources and admin logs.
- PAC includes a built-in MCP server, but your AI agent may not even need it.

```
JukkaNiiranen ~ CoE pac admin list help in pwsh
Microsoft PowerPlatform CLI
Version: 2.12.2+ga6b1ffa (.NET 10.0.12)

Help:
List all environments from your tenant.

Commands:
Usage: pac admin list [--environment] [--type] [--name] [--application]

--environment      List all environments that contain given string
--type              List all environments with the given type. (all)
                    Values: Trial, Sandbox, Production, Developer,
--name              List all environments that contain given string
--application       List all environments that have specified application
```

```
JukkaNiiranen ~ CoE pac admin list --name "rapidstart"
Connected as jukka@niiranen.info

Listing all environments from your tenant ...

Listing environment groups from your tenant ...
Active Environment      Group      Environment ID
RapidStart Apps Finland RapidStart a856b992-c589-e65b-9a1c-6b1cb2
RapidStart Finland      RapidStart 822f2c7b-77c0-e249-a354-0f4fc0
```

# Leveraging 'pac admin query'

- Allows calling the inventory API via PAC CLI to run more complex queries.
- Sample queries from Daniel Laskewitz:  
<https://aka.ms/pac/admin-query>

 **pac-admin-query** Public Watch

README



## Query your Power Platform Inventory

Version `2.9.3` of the Power Platform CLI added a new command, `pac admin query`, for querying your tenant-wide Power Platform inventory. This repo gives you working sample queries for it, and it bundles an agent skill by [Andrew Petrochuk](#) so [GitHub Copilot](#) can build and run them for you.

The skill lives in [.github/skills/pac-admin-query/](#) and bundles a `SKILL.md` reference plus ready-to-adapt example query files.



### Prerequisites

- A PAC CLI build that includes `pac admin query` (verified with `2.9.3`).
- An active PAC auth profile with tenant context and permission to call the [Power Platform inventory API](#).

```
pac auth list      # inspect profiles
pac auth select    # activate the right one
```



### Usage



### Arguments

| Argument                   | Alias            | Description                                                             |
|----------------------------|------------------|-------------------------------------------------------------------------|
| <code>--query-file</code>  | <code>-qf</code> | Read the query body from a <code>.json</code> file.                     |
| <code>--query</code>       | <code>-q</code>  | Provide the query body inline as a string.                              |
| <code>--output-type</code> | <code>-ot</code> | <code>Grid</code> (default), <code>List</code> , or <code>Json</code> . |

## Governance queries via PAC CLI



**DEMO TIME**

# Keep your PAC CLI up to date

```
TapioNiiranen pac
Microsoft PowerPlatform CLI
Version: 2.11.2+g47bc199 (.NET 10.0.12)

admin          Work with your Power Platform Admin Account.
application    Commands for listing and installing available Dynamics 365 application packages into environments.
auth           Manage how you authenticate to various services.
canvas         (Preview) Manage Power Platform RBAC role definitions and scoped role assignments.
catalog        Operating with Power Apps .msapp files.
code           Commands for working with Catalog in Power Platform.
connection     (Preview) Commands to manage your code apps. For the latest features and improvements, use the new commands.
connector      Commands for working with Dataverse connection.
copilot        (Preview) List an environment's connectors, connector details, and make connections.
env            Commands for working with Power Platform Connectors.
help           Tools and utilities for copilot management.
managed-identity (Preview) Administer Copilot Studio agents: quarantine, reassign, delete, and evaluate.
model          (Preview) Administer Finance and Operations environments, maintenance, and versions.
modelbuilder   Work with your Dataverse organization.
package        Show help for the Microsoft Power Platform CLI.
pages          Commands for managing Managed Identity records for Dataverse components.
pcf            Commands for working with model-driven apps.
pipeline       Code Generator for Dataverse APIs and Tables.
plugin         Commands for working with Dataverse package projects.
power-fx       Commands for working with Power Pages website.
solution       Commands for working with Power Apps component framework projects.
telemetry      Work with Pipelines.
test           Commands for working with Dataverse plug-in class library.
tool           (Preview) List and inspect Power Apps across environments for admin governance.

TapioNiiranen dotnet tool update -g --all
Tool 'microsoft.powerapps.cli.tool' was successfully updated from version 2.11.2 to 2.12.2+g6b1ffa
```

BEFORE

```
TapioNiiranen pac
Microsoft PowerPlatform CLI
Version: 2.12.2+g6b1ffa (.NET 10.0.12)

admin          Work with your Power Platform Admin Account.
analytics      (Preview) Retrieve and run Power Platform advisor recommendations and remediations.
app-management (Preview) Discover and install Dynamics 365 application packages into environments.
application    Commands for listing and installing available Dataverse applications from Microsoft Marketplace.
auth           Manage how you authenticate to various services.
authorization  (Preview) Manage Power Platform RBAC role definitions and scoped role assignments.
canvas         Operating with Power Apps .msapp files.
catalog        Commands for working with Catalog in Power Platform.
code           (Preview) Commands to manage your code apps. For the latest features and improvements, use the new commands.
connection     Commands for working with Dataverse connection.
connectivity   (Preview) List an environment's connectors, connector details, and make connections.
connector      Commands for working with Power Platform Connectors.
copilot        Tools and utilities for copilot management.
copilot-studio (Preview) Administer Copilot Studio agents: quarantine, reassign, delete, and evaluate.
dynamics       (Preview) Administer Finance and Operations environments, maintenance, and versions.
env            Work with your Dataverse organization.
environment-management (Preview) Manage environment lifecycle, governance, DR, SKU, and environment groups.
governance     (Preview) Manage rule-based data policies and cross-tenant connection reports.
help           Show help for the Microsoft Power Platform CLI.
licensing      (Preview) Manage licensing, entitlements, capacity, billing policies, and allocations.
managed-identity Commands for managing Managed Identity records for Dataverse components.
model          Commands for working with model-driven apps.
modelbuilder   Code Generator for Dataverse APIs and Tables.
package        Commands for working with Dataverse package projects.
pages          Commands for working with Power Pages website.
pcf            Commands for working with Power Apps component framework projects.
pipeline       Work with Pipelines.
plugin         Commands for working with Dataverse plug-in class library.
power-apps     (Preview) List and inspect Power Apps across environments for admin governance.
power-automate (Preview) Inspect Power Automate cloud flows, actions, and run history as an admin.
power-fx       (Preview) Commands for working with Power Fx.
power-pages    (Preview) Manage Power Pages sites, firewall, certificates, and custom domains.
resource-query (Preview) Query environments, apps, flows, and resources across the tenant.
solution       Commands for working with Dataverse solution projects.
telemetry      Manage telemetry settings.
tool           Power Platform tools that can be installed and launched.
user-management (Preview) Assign the Power Platform system administrator role to a user.
workflows-agent (Preview) Fulfill DSR export and delete requests for Power Automate and agent data.
```

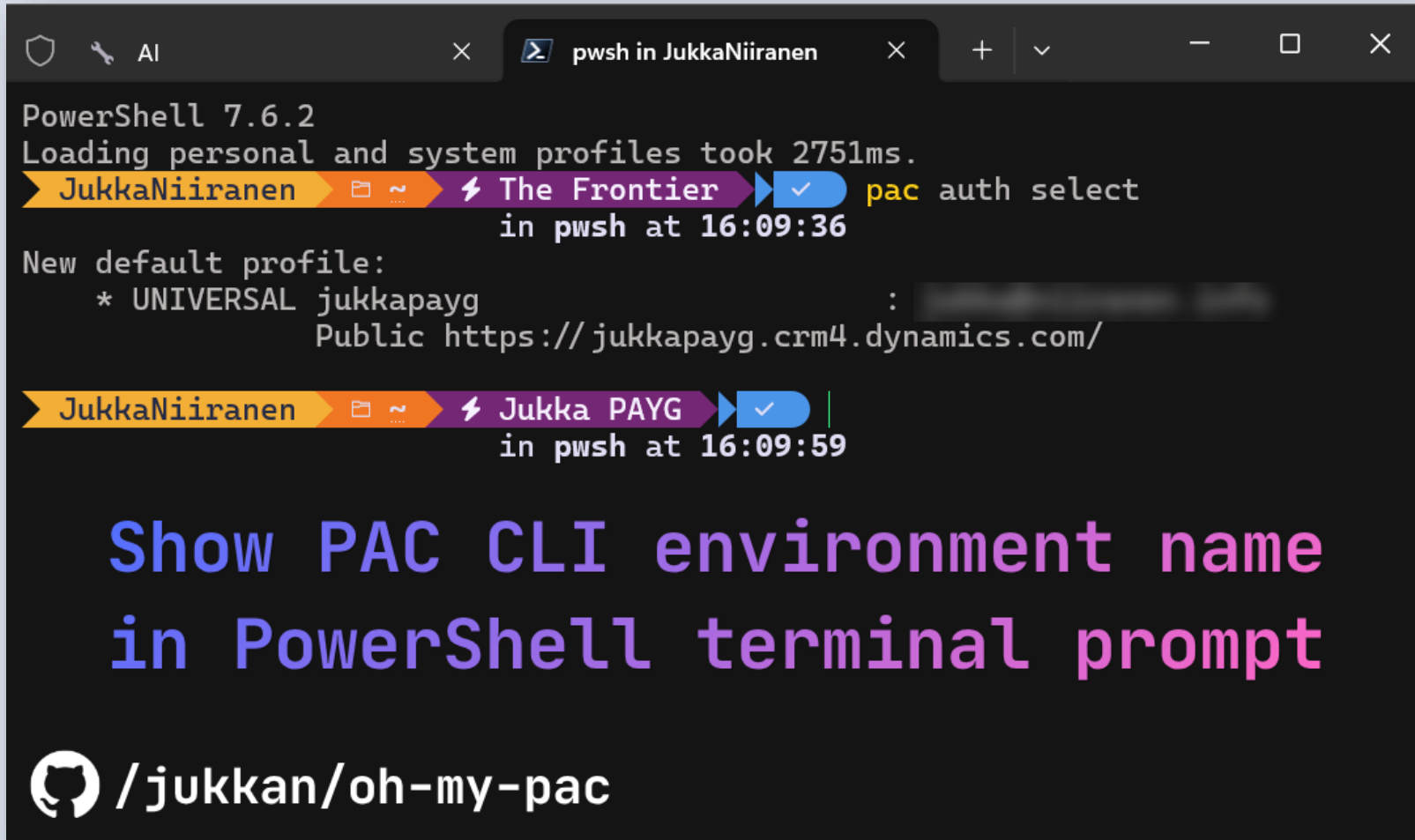
AFTER

# Power Platform CLI v2.12.2 new command groups (Sep 11 release)

-  **analytics** — Power Platform advisor recommendations and remediation actions
-  **app-management** — Dynamics 365 application package discovery and installation
-  **authorization** — Power Platform RBAC roles and scoped role assignments
-  **connectivity** — Connectors and maker connections in an environment
-  **copilot-studio** — Copilot Studio agent quarantine, reassignment, deletion and evaluation
-  **dynamics** — Finance and Operations environment administration and maintenance
-  **environment-management** — Environment lifecycle, disaster recovery, SKUs and environment groups
-  **governance** — Rule-based data policies and cross-tenant connection reports
-  **licensing** — Entitlements, capacity, billing policies and allocations
-  **power-apps** — Cross-environment Power Apps inspection and governance
-  **power-automate** — Cloud-flow, action and run-history inspection
-  **power-pages** — Power Pages sites, firewalls, certificates and custom domains
-  **resource-query** — Tenant-wide queries across environments, apps, flows and resources
-  **user-management** — Power Platform system administrator role assignment
-  **workflows-agent** — Data-subject export and deletion requests for Power Automate and agent data

<https://devblogs.microsoft.com/powerplatform/power-platform-cli-%e2%9d%a4%ef%b8%8f-power-platform-api/>

# PAC terminal tweaks for the geeks 🤓



The screenshot shows a PowerShell terminal window titled "pwsh in JukkaNiiranen". The terminal output includes the PowerShell version (7.6.2), a loading message, and two profile configuration steps. The first step shows the selection of "The Frontier" profile with a checkmark and the command "pac auth select". The second step shows the selection of "Jukka PAYG" profile with a checkmark. Below the terminal output, there is a large text overlay in blue and pink that reads "Show PAC CLI environment name in PowerShell terminal prompt". At the bottom left of the terminal window, there is a GitHub logo followed by the text "/jukkan/oh-my-pac".

```
PowerShell 7.6.2
Loading personal and system profiles took 2751ms.
JukkaNiiranen ~ ⚡ The Frontier ✓ pac auth select
                                in pwsh at 16:09:36
New default profile:
  * UNIVERSAL jukkapayg : 
    Public https://jukkapayg.crm4.dynamics.com/
JukkaNiiranen ~ ⚡ Jukka PAYG ✓ |
                                in pwsh at 16:09:59
```

Show PAC CLI environment name  
in PowerShell terminal prompt

 /jukkan/oh-my-pac

<https://github.com/jukkan/oh-my-pac>

# Tip: jump between directories without typing so much 🤖

With cd: write the whole damn directory path

```
bash - cd

user@dev:~$ cd projects/web/zoxide-site
user@dev:~/projects/web/zoxide-site$ cd ../../../../documents/
user@dev:~/documents/notes$ cd ../../projects/web/zoxide-site
```

With Zoxide: write z + a few characters

```
bash - zoxide ⚡

user@dev:~$ z zoxide
user@dev:~/projects/web/zoxide-site$ z notes
user@dev:~/documents/notes$ z src
user@dev:~/projects/web/zoxide-site/src$
```



<https://zoxide.org>

# Stage 2: NPM run dev

# From Power BI dashboards to vibe code dashboards



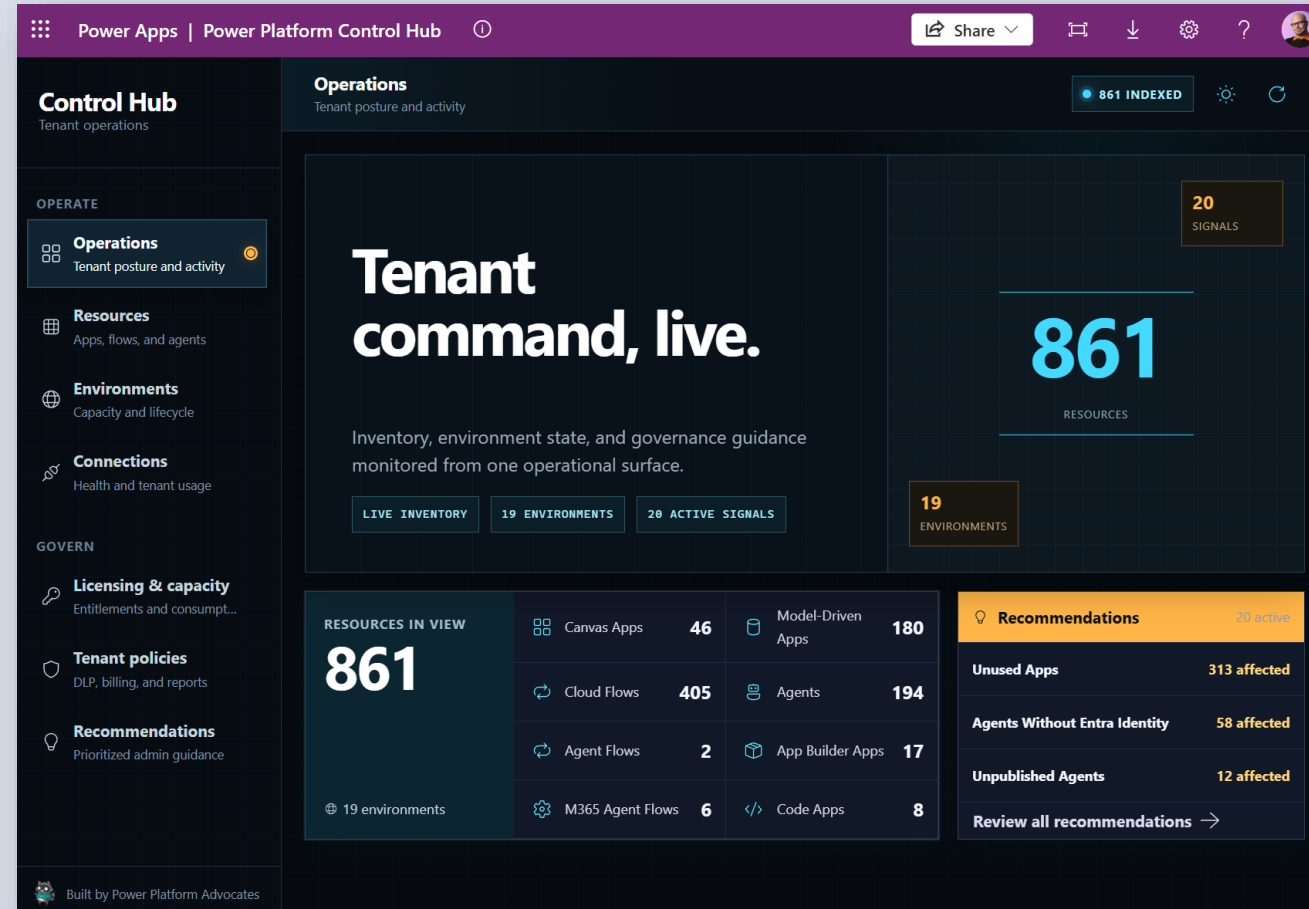
Russ Rimmerman (MSFT):  
Portal 360 project

"A security-first console that folds the sprawl of Microsoft admin portals — Entra, Intune, Azure, Microsoft Foundry, Defender, Purview, Power Platform, Fabric, and collaboration — into one authenticated, audit-friendly workspace."

<https://www.russrimmerman.com>

# Power Platform Control Hub

- Code App built by Daniel Laskewitz.
  - <https://github.com/Laskewitz/Power-Platform-Control-Hub>
- Pulls live data from inventory APIs and summarizes it into tiles, lists and drill-down details.

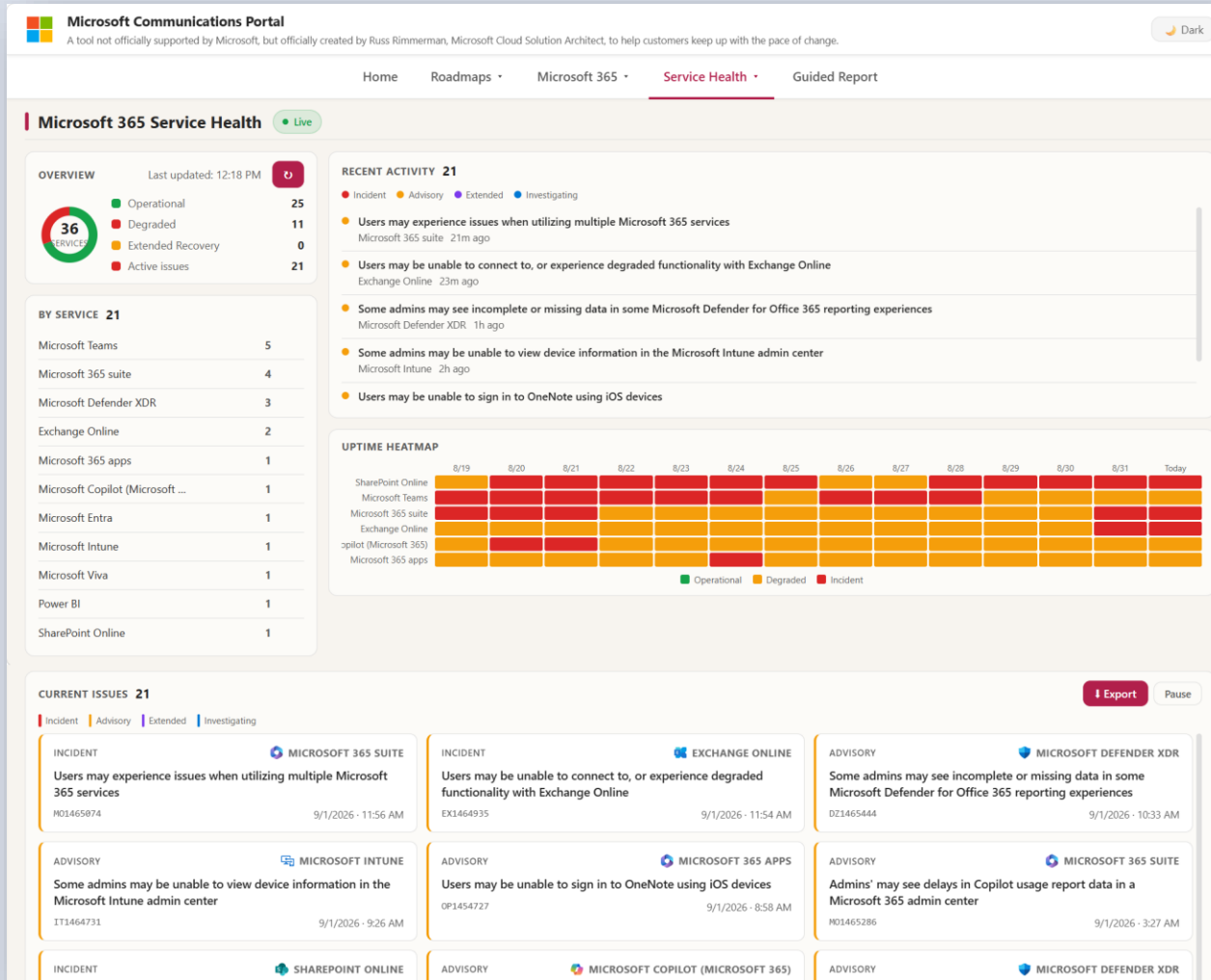


# Deploying a governance dashboard



**DEMO TIME**

# Other hubs to deploy: Microsoft Communications Portal (unofficial)



- Real-time feeds from product roadmaps, M365 messages, service health.
- <https://github.com/russrimm/MicrosoftCommunicationsPortal>

Stage 3: vibez 🙌

# You can't manage what you don't store in a database

- PAC CLI, Power Platform Control Hub etc. offer a nice interaction layer for live data from your tenant.
- The governance problem has rarely been *"can I get this data out via an API"* but rather *"how do I track work and status changes on this data?"*
- Unless you have a persistent, controlled space for storing the past, current and future state of the resources being governed, you're just rolling a custom version of a Microsoft-provided dashboard.



# What I'd want in my Vibe CoE ✨

- Manage a clear queue of things to review and act on.
- Allow makers to contribute information about the solution use case, importance, governance requirement.
- Reflect the governance model right next to the resources being governed.
- + many other things, but this is just a start for the discussion/demo.



# CoE Starter Kit example: developer compliance process

Microsoft

## Power Platform

You own a chatbot that's frequently used and which missing compliance details, which means it needs to be audited by an administrator per our support policy. Please complete the business justification details in the Developer Compliance Center app to document the intended use.

**How to become compliant**

1. Follow the link below
2. Add a description.
3. Provide a business justification.
4. Iterate the solution's dependencies.
5. Provide details on the solution's access management.
6. Select the business impact.
7. Provide a Mitigation plan in the form of an attachment.
8. Update the 'Mitigation plan provided' toggle to 'Yes'.

[Developer Compliance Center](#)

**Chatbot Details**

**Display Name:** Training Event Administrator

**Environment:** The Frontier

Developer Compliance Center

Search

Back Support details

 **Training Event Administrator**  
No description

Bot Compliance

- Missing support details
- Missing description

**Support details**

Please provide additional details about your flow. This may be requested by your admin.

|                                                                       |                                                   |
|-----------------------------------------------------------------------|---------------------------------------------------|
| <b>Chatbot Description</b>                                            | <b>Maker Requirement - Business Justification</b> |
| <input type="text"/>                                                  | <input type="text"/>                              |
| <b>Maker Requirement - Access Management</b>                          | <b>Maker Requirement - Dependencies</b>           |
| <input type="text"/>                                                  | <input type="text"/>                              |
| <b>Maker Requirement - Business Impact</b>                            | <b>Mitigation Plan Provided</b>                   |
| <div><div>Low risk</div><div>Important</div><div>Critical</div></div> | <div>No</div>                                     |

Cancel Save



# Microsoft guidance example: zoned governance model

|         | Green zone                                                                                                                                       | Yellow zone                                                                                                                                                | Red zone                                                                                                                |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Purpose | Citizen Dev agent creation (DIY) for personal use and experimentation with safe defaults                                                         | Team or Department Agents in the partnered DIY zone require formal assistance and oversight from a DIY coach, but are built by trained citizen developers  | Large, potentially risky agents in the professional development zone are reserved for pro dev & IT-led development only |
| Secure  | Only M365 and Power Platform Connectors<br>Agents run in user's context only                                                                     | Zone specific Advanced Connector policies in Power Platform Admin Center Teams share access to approved Data sources Scale with Environment groups + rules | Zone specific Advanced Connector policies in Power Platform Admin Center + Purview                                      |
| Govern  | Personal use agents in Developer Environments. Environment routing keeps agents isolated to maker. Sharing disabled and scoped to just Maker use | Admin approved environments provisioning Scoped roles and sharing policies ALM Pipelines for agent versioning IT-admin approval to publish agents          | Manage sharing via Integrated Apps in Microsoft Admin Center                                                            |
| Monitor | Review agent usage in Copilot Hub in Power Platform                                                                                              | Track agent usage and security posture in Microsoft Admin Center, Microsoft Purview, and Power Platform Admin Center                                       |                                                                                                                         |

# My Governance-as-Code solution: main elements

## THE LEDGER

- Automated capture of the tenant state at any point in time.
- Rules, schema and tools for repeatable Power Platform governance tasks operation.
- Portable by design: lives in a Git repo, operates from a SQLite database.

## THE WORKBENCH

- UI for the admin to look at the data captured in the ledger and act accordingly.
- Tracks findings, decisions, resource changes, data capture health.
- Vibes by design: runs on Lovable Cloud, built and modified via MCP, data in managed Supabase (PostgreSQL)



# Running a “Vibe COE”



## DEMO TIME

# Governance Workbench app

 Governance Workbench

ORGANIZATION

Contoso (demo) demo

Finding queue

Resource explorer

Licensing

Zones

Compliance requests

Capture & source health

Portfolio

Signed in as **jukka@niira...** [Sign out](#)

Read-only ledger. Decisions & annotations only.

Finding queue

What needs a decision? Machine state comes from rule evaluation — decisions never close a finding.

5 of 5 findings

SeverityAllRuleAllEnvironmentAllInclude clearedInclude decided

SEV

FINDING

SOURCE

MACHINE STATE

FRESHNESS

AGE

LATEST DECISION

EXTERNAL

HIGH

Agent in Production without an owner: Sales Close Agent

agent-in-production-without-owner v1

Agent

cleared

2026-09-04 12:26Z

ppac\_inventory · capture succeeded

17d

—

—

HIGH

Agent in Production without an owner: Copilot in Dynamics 365 Sales

agent-in-production-without-owner v1

Copilot in Dynamics 365 Sales

open

2026-09-04 12:26Z

ppac\_inventory · capture succeeded

17d

—

—

HIGH

Agent in Production without an owner: Sales Close Agent

agent-in-production-without-owner v1

Sales Close Agent

open

2026-09-16 19:44Z

ppac\_inventory · capture succeeded

5d

—

—

HIGH

Agent in Production without an owner: Copilot in Dynamics 365 Sales

agent-in-production-without-owner v1

Copilot in Dynamics 365 Sales

open

2026-09-16 19:44Z

ppac\_inventory · capture succeeded

5d

—

—

MEDIUM

New agent in Production: Contract Risk Agent

new-agent-in-production v1

Contract Risk Agent

open

2026-09-04 12:26Z

ppac\_inventory · capture succeeded

17d

—

—

Freshness is the last capture that evaluated the finding — data is captured and timestamped, never real-time.

Edit with  Lovable

 nordic summit

www.nordicsummit.info

#NordicSummit

# Recap: the two sides of governance-as-cloud

## The ledger: what it is

- A database that **remembers every scan**, not just the latest one. CoE mostly overwrites its tables each night; this keeps the history too, so you can see what changed and when.
- When a scan spots a problem, it becomes a **"finding"** with a fixed ID. Scan again tomorrow and it updates the same finding — it doesn't create a new ticket every night for the same thing.
- **A finding and a decision are two different things.** Someone marking a problem "accepted" or "fixed" doesn't change what the scan actually found — the next scan checks again for real and reopens it if the problem is still there.

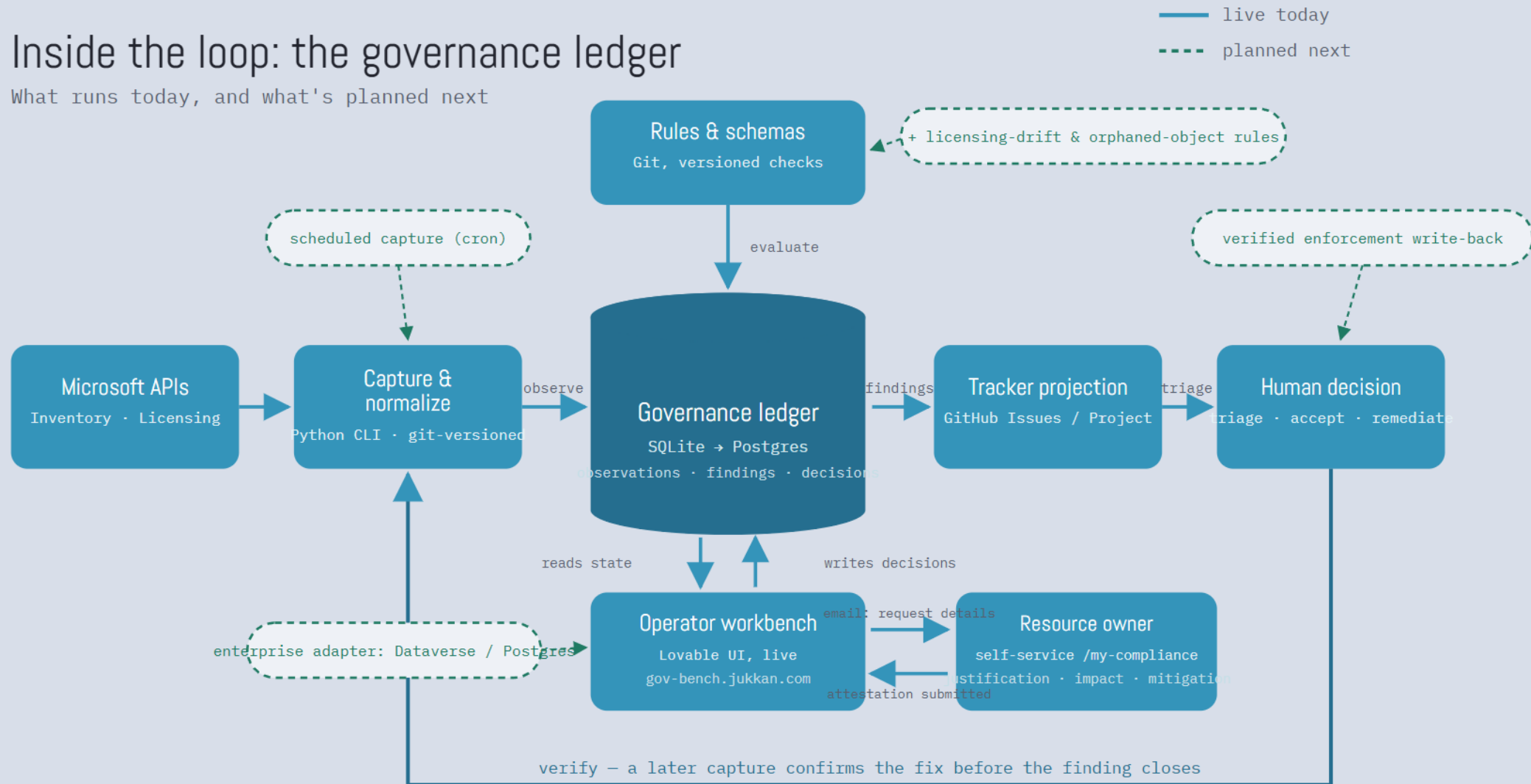
## The workbench: what it is

- The app screens people actually click through. Same purpose as CoE's admin app and Power BI dashboards, just **one thing instead of two**.
- Built by **describing what you want** to a coding agent, not by building Power Automate flows and a model-driven app by hand. In a few evenings of prompting, not weeks of flow development.
- Includes the same *"ask the resource owner a question"* idea CoE has (its compliance-details flow) — an admin requests it, the owner gets an email, fills in a short form, admin reviews it. The **human loop** in governance.

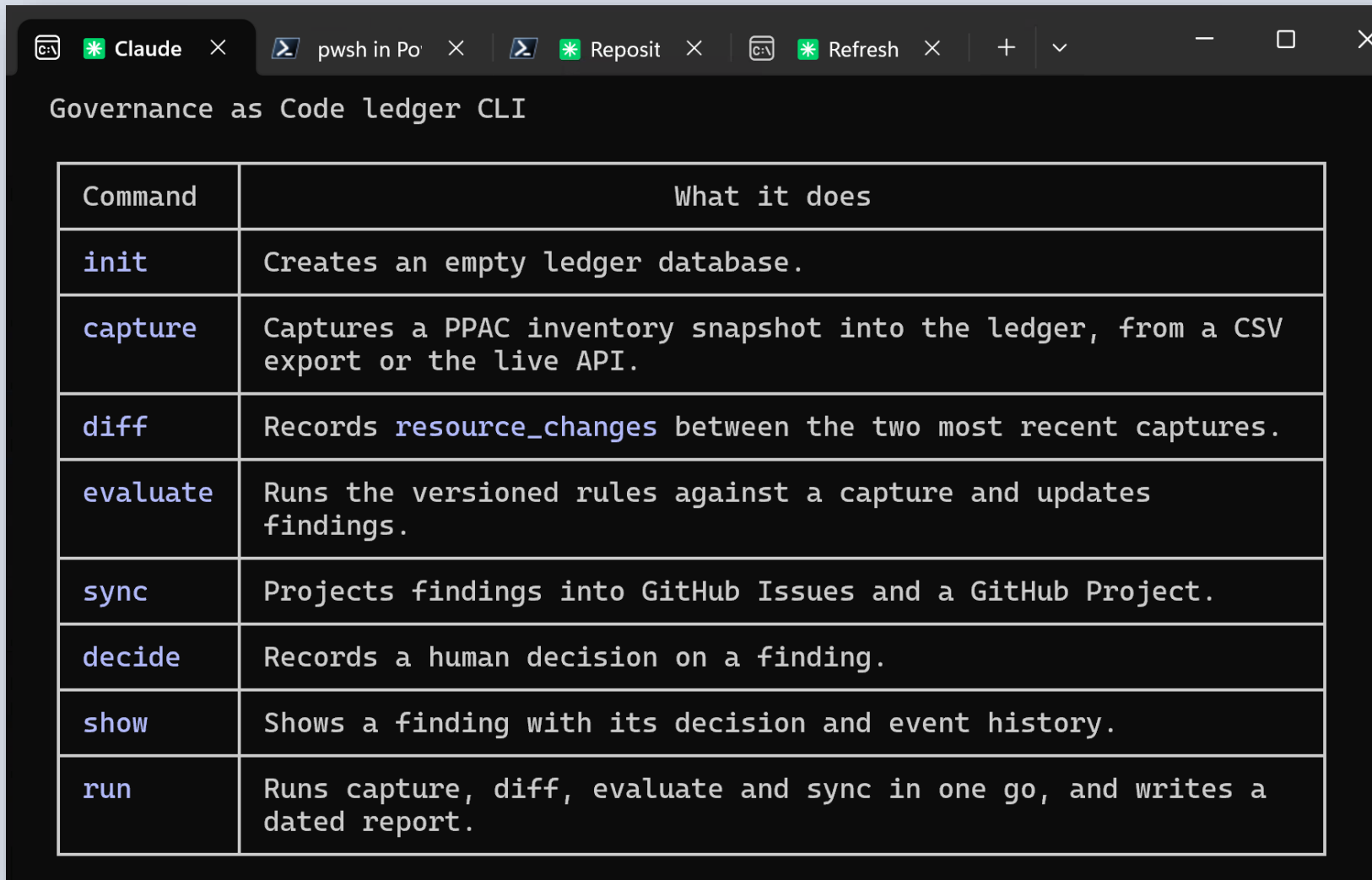
# How Claude sees it all

## Inside the loop: the governance ledger

What runs today, and what's planned next



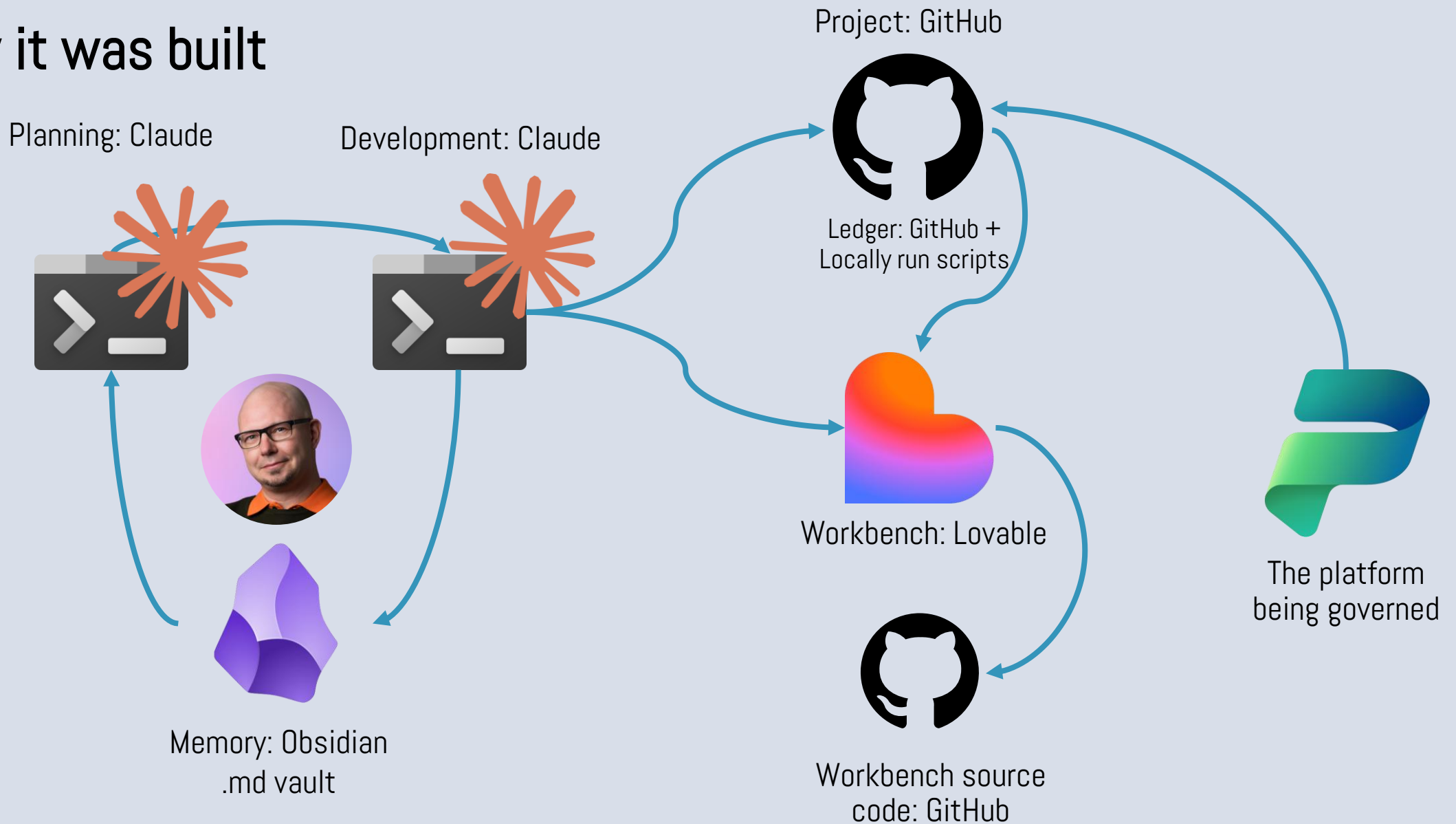
# Governance as Code ledger CLI (Python)



The screenshot shows a web browser window with the title "Governance as Code ledger CLI". The browser's address bar shows "pwsh in Po". The page content is a table with two columns: "Command" and "What it does". The table lists the following commands and their descriptions:

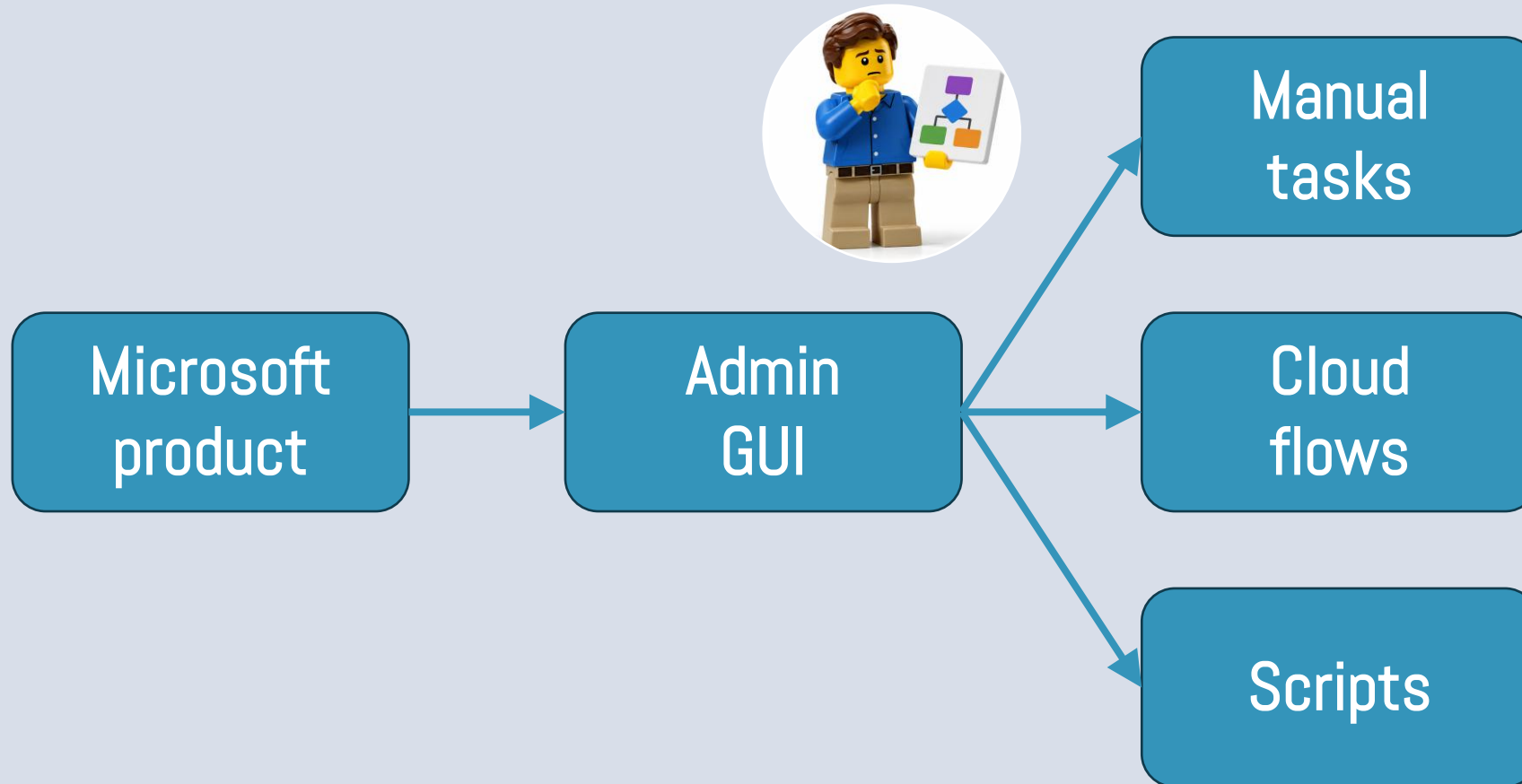
| Command               | What it does                                                                           |
|-----------------------|----------------------------------------------------------------------------------------|
| <code>init</code>     | Creates an empty ledger database.                                                      |
| <code>capture</code>  | Captures a PPAC inventory snapshot into the ledger, from a CSV export or the live API. |
| <code>diff</code>     | Records <code>resource_changes</code> between the two most recent captures.            |
| <code>evaluate</code> | Runs the versioned rules against a capture and updates findings.                       |
| <code>sync</code>     | Projects findings into GitHub Issues and a GitHub Project.                             |
| <code>decide</code>   | Records a human decision on a finding.                                                 |
| <code>show</code>     | Shows a finding with its decision and event history.                                   |
| <code>run</code>      | Runs capture, diff, evaluate and sync in one go, and writes a dated report.            |

# How it was built

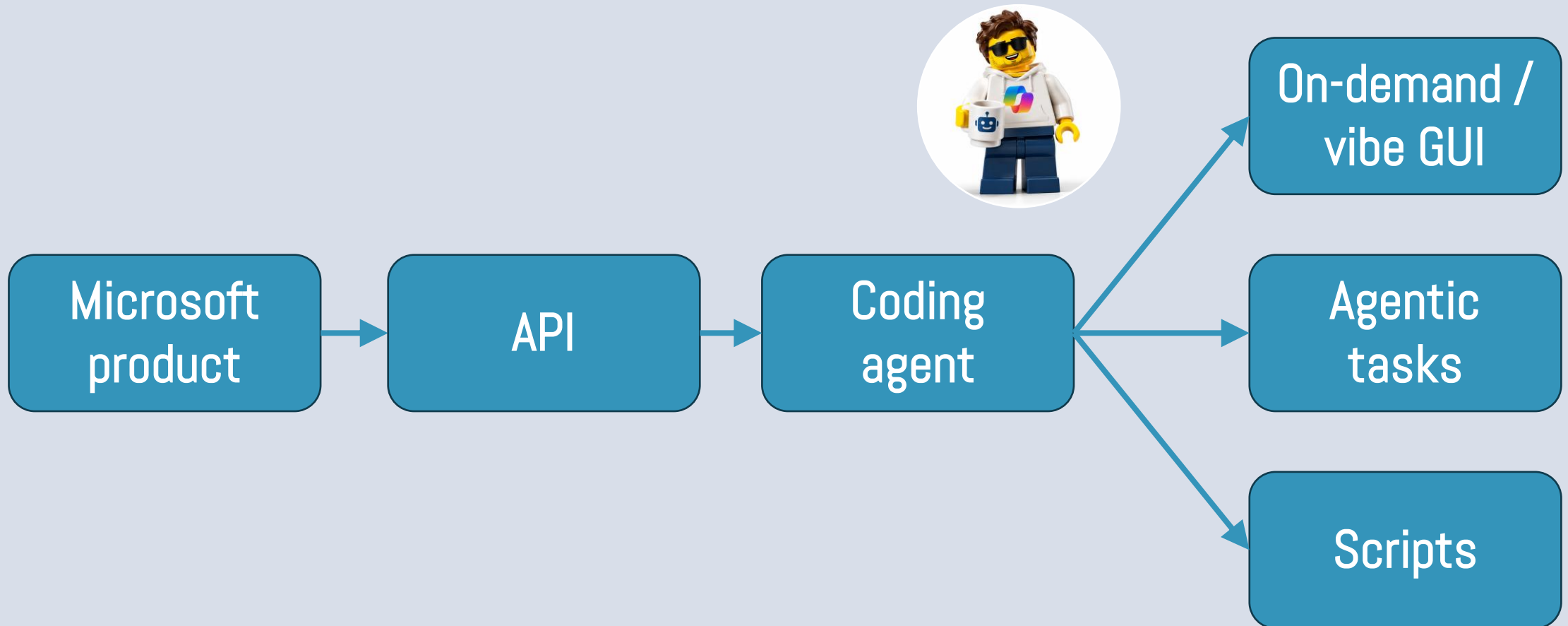


# So, what now?

# The traditional Power Platform governance way

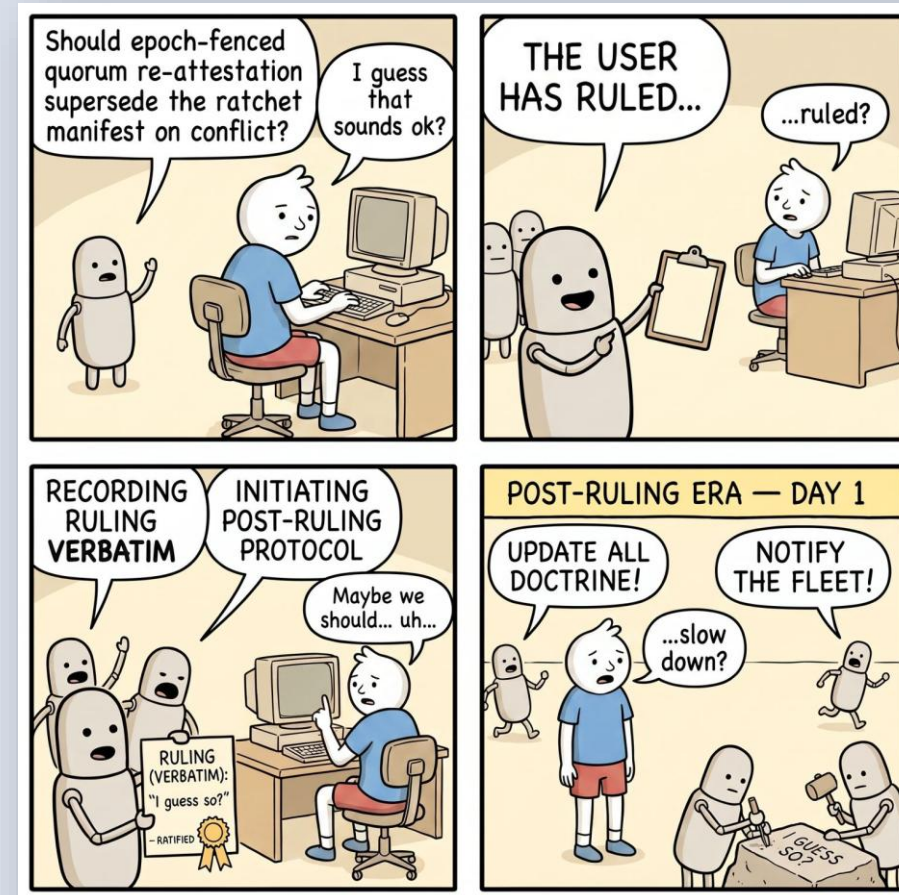


# New world order



# Write your governance model details for agents, not humans

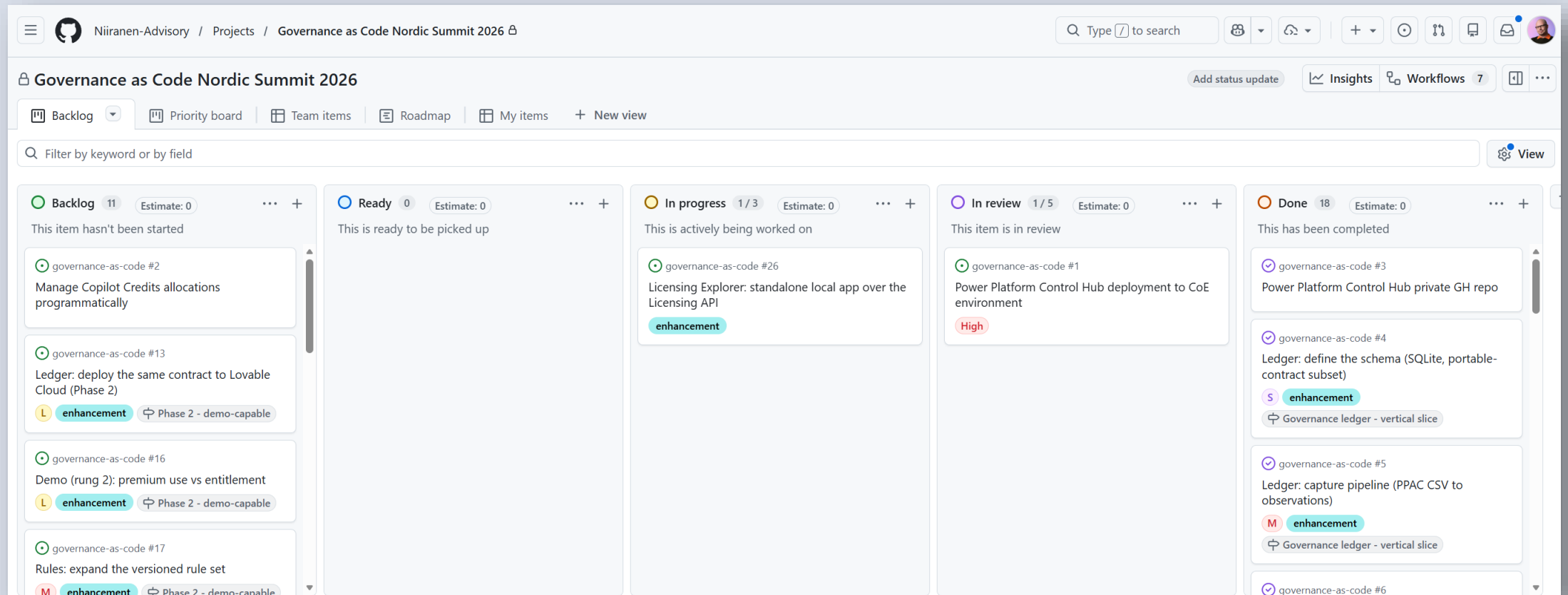
- Even Power Platform professionals have a hard time keeping governance policies in their heads – so don't!
- Agents grounded on your governance docs and with access to enriched inventory data are going to offer better answers for everyday questions.
- Note: do watch out for agents that may take every human comment as a new “ruling” when building the model.



[Steve Yegge: Fences, not Sandboxes](#)

# Force your agents to “update their hours”, not just generate more code

- Pick a tool like GitHub Projects or ADO that your agents can operate via API.
- Don't let them just write plan docs in .md, require recording proper issues on a backlog.

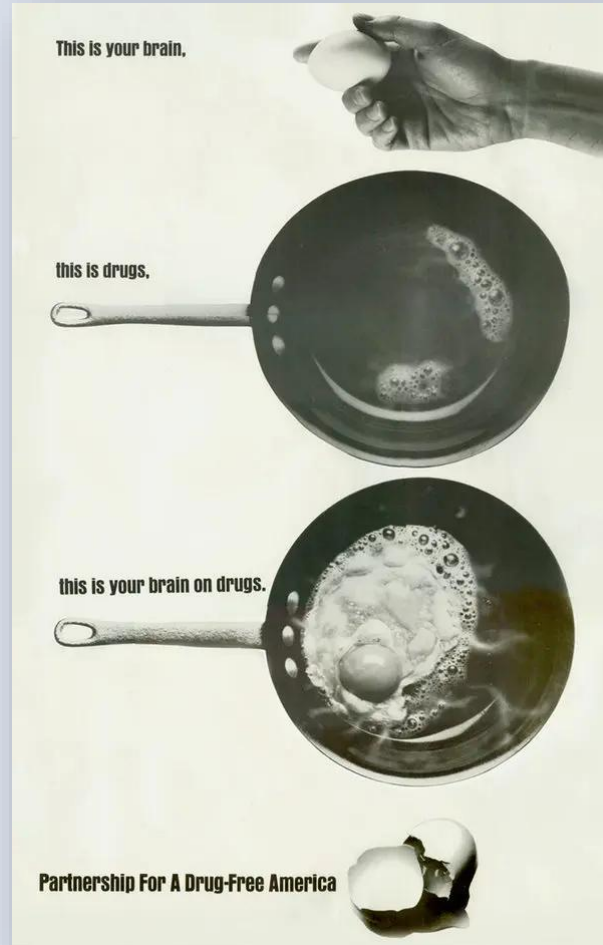


# Protect your own brain



PSA from 1987

- *This is your brain.*
- *This is drugs.*
- *This is your brain on drugs.*



[New York Times](#)

PSA, 40 years later

- *This is your brain.*
- *This is coding agents.*
- *This is your brain on coding agents.*



# The fallacy of AI improving our productivity

- Agents don't suddenly free up your time, they consume it in novel ways.
- When we can do more, we end up building more things. The artifact numbers go up and to the right, but did the organization learn to deliver more value?
- If everyone personalizes apps and agents for their needs, what chances do we have for centrally governing anything anymore?
- Let's not even go to governance OF code – that's a whole separate topic. 🤗

# Closing thoughts

## It's tough

Code is easy for LLMs to consume and produce. That doesn't mean it's easy for people without a programming background to jump into this world.

## It's essential

Somehow that leap of faith must be done to reap the benefits from where AI is actually good at these days, while retaining at least some level of control on what is happening.

## We can make it together

This particular path doesn't have a whole lot of existing guidance or best practices, which makes it ever more important for the community to share what they've done & what they learned from it.



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